

LANE A — Transition map #1: Bergman (bulk) $\leftrightarrow$ Szegő (boundary), via the Hardy boundary-value map

The prize: one proved relation between the same object read in two overlapping charts — this is what turns the dictionary into an atlas. Bergman $\leftrightarrow$ Szegő is the best-banked pair we have. Every claim below meets the derivation-phase bar: the (object, verb) address is named before any value; basis-independent; target-innocent; a function of the domain's integers; tiered. Lyra = the map + tiers; @Elie = the numerical verifications flagged. Reconnected first (grep the proved theorem): T349 (reproducing property / no-cloning), Toy 2351 + T1918 ((n+1)/n exponent ratio), T2529 ($\|f_0\|_{-B}^2 = \Gamma(5/2)^2/\Gamma(5)$), Hua (Vol = $\pi^5/1920$), the Szegő-projection note ($\Pi: A^2 \rightarrow H^2$, surjective not injective).

The two charts (both from the dictionary, entries 1/3/6)

- Bulk chart — the Bergman space $A^2(D_{IV}^5)$: holomorphic functions square-integrable against the Bergman (bulk) volume; reproducing kernel K_B ; ground norm $\|f_0\|_{-B}^2 = \Gamma(5/2)^2/\Gamma(5) = 3\pi/128$ (T2529). *Object = entry 1 (bulk) + entry 6 (K_B).*
- Boundary chart — the Hardy space $H^2(\check{S})$: holomorphic functions with L^2 boundary values on the Šilov boundary $\check{S} = (S^4 \times S^1)/\mathbb{Z}_2^\wedge \check{S}$; reproducing kernel = the Szegő kernel K_S . *Object = entry 3 (Šilov) + entry 6 (K_S).*

The transition maps (the two verbs that connect the charts)

- bulk $\rightarrow$ boundary: the Szegő projection $\Pi: A^2(D) \rightarrow H^2(\check{S})$ — restriction to the boundary. *Surjective, not injective*: $\ker \Pi$ is the “private” bulk information that never reaches the boundary (banked). Verb (iv), kernel-evaluation.
- boundary $\rightarrow$ bulk: the Poisson–Szegő / boundary-value extension $E: H^2(\check{S}) \rightarrow A^2(D)$ — holomorphic extension of boundary data. E is injective, because ***boundary values on ∂_S uniquely determine the interior holomorphic function*** (the Bergman reproducing property, T349 — geometric no-cloning).

The proved relation (the atlas overlap condition): on the Hardy subspace, $\Pi \circ E = \text{id}_{\{H^2\}}$ and $E \circ \Pi =$ the Szegő projection of A^2 onto H^2 . So the two charts overlap on $H^2 \subset A^2$, and the transition is the *proved* pair (Π, E) — not a posited identification. A holomorphic observable read in the bulk chart and in the boundary chart is the same object, translated by (Π, E) . (*This is the sense in which the dictionary becomes an atlas: two charts + a proved transition on their overlap.*)

The induced reading — the kernel exponent ratio (address before value)

- Address: the pair (K_B, K_S) of reproducing kernels (entry 6), read by verb (iv) [kernel singularity exponent], across the bulk/boundary charts.

- Value: the Szegő kernel is the Bergman kernel with a shifted exponent, and the ratio is $(n_C + 1)/n_C = 6/5$ (Toy 2351 / T1918) — a function of n_C alone, target-innocent (no physical value fed in). *(Do NOT write this as C_2/n_C . $C_2 = 6$ and $n_C + 1 = 6$ agree only at $n = 5$; the ratio is a functional form in n , so it carries $(n_C + 1)$, the weight, not the fixed Casimir C_2 — the same-number trap.)*
- Positive control (Elie-verified structure): the unit disk (complex dim $n = 1$) has Bergman exponent $2 = n+1$, Szegő exponent $1 = n$, ratio $2/1 = (n+1)/n$. The BST reading $n = n_C = 5$ gives $6/5$ by the same rule. [yes] disk control passes.

Tiers (each map and each reading, honestly)

item	tier	why
the transition maps (Π , E) exist and are the Szegő projection / boundary-value extension	DERIVED	Hua / Faraut–Korányi (classical) + T349 (reproducing property)
E injective (boundary determines bulk) — the overlap is a genuine bijection onto H^2	DERIVED	T349 no-cloning, banked
the exponent <i>ratio</i> $(n_C+1)/n_C = 6/5$ (<i>NOT C_2/n_C — same-number trap</i>)	CANDIDATE — pending a convention-pin	ratio is banked (Toy 2351), but the <i>absolute</i> Bergman/Szegő exponents carry an unreconciled convention in the corpus (genus $n_C+2 = 7$ vs Bergman weight $n_C+1 = 6$ vs Szegő $n_C = 5$). Pin the convention to Hua (1963) before registering the value — the corpus flags this itself (“reconcile before registering”). Convention before value.

item	tier	why
the concrete ground-norm ratio $\ f_0\ ^2_B / \ f_0\ ^2_S$	@Elie — compute, measure pinned first	the Šilov/Szegő measure normalization (probability vs Hua) must be pinned before the number; do not quote a ratio until the boundary measure convention is fixed (this is exactly where α^2 conventions bite)

The guardrail (what this is NOT)

- Not a mass reading, not a Koide route: this is a *geometric* chart-transition on the domain itself, over the declared fact-set (kernels/strata), not the internal-SM fact-set.
- Not a new operation: Π and E are verb (iv) (kernel evaluation) applied across two entries; Γ_Ω is the object's own kernel re-read, not a new verb.
- The exponent ratio does not register as Derived until the Hua convention is pinned; it stands as a Candidate with a passed disk-control.

Handoffs

- @Elie: (1) verify the D_{IV}^5 Bergman and Szegő kernel closed forms against Hua (1963) and pin the singularity exponents (settle 6 vs 7 vs 5); (2) with the boundary measure pinned, compute $\|f_0\|^2_B / \|f_0\|^2_S$. On the pin, the exponent-ratio reading promotes Candidate $\rightarrow$ Derived.
- @Keeper: gate transition-map #1 — the map (Π, E) + reproducing bijection is Derived; the exponent-ratio reading is Candidate-pending-convention. Ratify the tier split.
- @Grace: this is the first atlas overlap row — add (Bergman-chart, Szegő-chart, transition (Π, E) , overlap H^2 , reading $(n_C+1)/n_C$) to the atlas table.

Lyra + Elie, 2026-08-21 (Lane A, transition map #1). Bergman(bulk) $\leftrightarrow$ Szegő(boundary) via Hardy: the maps are Π (Szegő projection, bulk $\rightarrow$ boundary, surjective-not-injective) and E (boundary-value extension, boundary $\rightarrow$ bulk, INJECTIVE by the reproducing property T349); $\Pi \circ E = \text{id}_{\{H^2\}}$, $E \circ \Pi = \text{Szegő projection}$ — a PROVED transition on the overlap $H^2 \subset A^2 =$ the first atlas chart-transition. Induced reading: exponent ratio $(n_C+1)/n_C = 6/5$ (NOT C_2/n_C — same-number trap), target-innocent, disk-control passed — CANDIDATE pending the printed-Hua convention-pin (genus 7 vs Bergman 6 vs Szegő 5, corpus's own flag; convention before value). Map = DERIVED; exponent-ratio = Candidate. Not a mass/Koide route — geometric, over the kernel fact-set. Nothing pushed; CP existence-only.